higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATED GC

NATIONAL CERTIFICATE

AUGUST EXAMINATION

INDUSTRIAL INSTRUMENTS N5

28 JULY 2015

This marking guideline consists of 11 pages.

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SECTION A

QUESTION 1

1.1 Impact pressure developed:

$h = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$ Where $V_2 = 0$

Therefore ∴ $h = \frac{-V_1^2}{2g}$

The pressure inceases by = $\frac{V_1^2}{2g}$

Increase in head h = $\frac{V_1^2}{2g}$

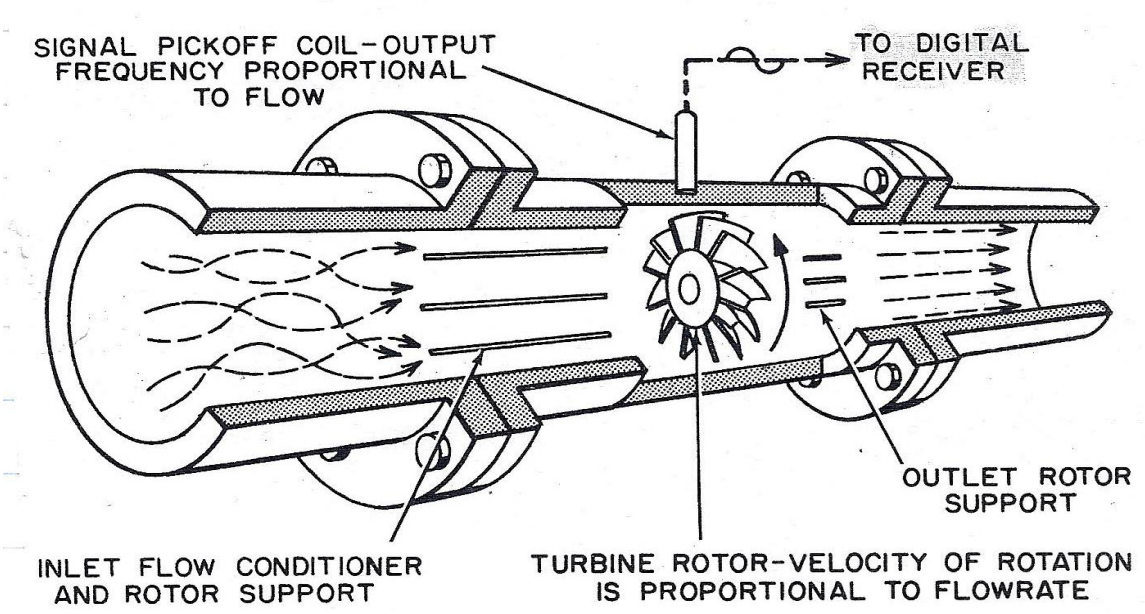
Or $V_1^2 = 2gh$

thus V1 = $\sqrt{2gh}$

Pitot tube coefficient is introduced $V_1 = C\sqrt{2gh}$ (5)

TED GO

1.2



The turbine flowmeter provides a frequency output signal that varies linearly with a volumetric flow rate over specified flow ranges with fluids of given kinematic viscosity.

The relationship between rotor speed and volumetric flow rate is linear within given limits of flow rate and fluid viscosity.

The pick-off assembly transduces the rotor velocity to an equivalent frequency signal. In this system the meter housing must be non-magnetic. The rotor must be of permeable material. The pick-off assembly consists of a small powerful permanent magnet and a coil winding. The field of the magnet is influenced by the moving blades of the rotor which are of permeable material.

As a rotor blade passes through the field of the magnet, it provides an easier path for the field and the field distorts thus moving across the coil winding. The relative motion between the magnetic field and the coil winding generates an AC voltage, the frequency of which is proportional to flow rate.

A turbine flowmeter is to develop a precisely known number of pulses for a given volume measured. (10)

1.3 No. The relative motion between the magnetic field and the coil winding generates an AC voltage, the frequency of which is proportional to flow rate. Stated another way, the meter generates a known frequency for a given flow rate. (2)
[17]

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QUESTION 2

2.1

. $N = \dfrac{Q\sqrt{p}}{0{,}01252 \times D^2 \times \sqrt{h}}$

$= \dfrac{738\sqrt{6{,}408}}{0{,}01252 \times (250)^2\sqrt{102}}$

$= \underline{0{,}23639}$

. N = CME

N = 0,23639

then m = 0,36 (graph)

and C = 0,608 (graph)

. $Rd = \dfrac{Qp}{15{,}8 \times \mu \times D \times \sqrt{m}}$

$= \dfrac{7{,}38 \times 6{,}408}{15{,}8 \times (1{,}28 \times 10\ ) \times 250 \times \sqrt{0{,}36}}$

$= \underline{15589{,}082}$

Zr = 1,002 (graph)

Zd = 1,002 (graph)

Z = Zr x Zd = 1,001

$$E = \frac{1}{\sqrt{1 - m^2}}$$

$$= \frac{1}{\sqrt{1 - (0,36)^2}}$$

= 1,07186

$$h/Pa = \frac{kPa \times 27,2}{kPa}$$

$$= \frac{1 \times 27,2}{\frac{381}{102}}$$

= 0,2589

. Є = 0,9966 (graph)

. Q = 0,01252 CZE Є d^2 √h/p

$$d^2 = \frac{Q}{0,01252\ CZE\ Є\ \sqrt{h/p}}$$

$$= \frac{738 \times \sqrt{6,408}}{0,01252 \times 0,608 \times 1,002 \times 0,9966 \times 1,07186 \times \sqrt{102}}$$

= 149,5 mm (11)

2.2 M = 0,36
Pressure loss = 55% (2)
[13]

NATED GO

QUESTION 3

3.1 Q θ
= 8/15 Tan $2\sqrt{}$ 2gH
= 0,53 x 0,577$\sqrt{}$ 2gH
= 0,53 x 0,577 x $\sqrt{}$2 X 10 x (0,6)
= 0,3813 m²/sec (4)

3.2
- Rectangular weir
- V-Notch weir
- Trapezoidel/Cipoletti weir
- Parabola weir (4)

[8]

TOTAL SECTION A: 38

SECTION B

QUESTION 4

4.1 pH of a soluction is a measure of the acidity or alkalinity of that soluction. (2)

4.2 The function of a reference electrode is to complete the electrical circuit with the glass membrane electrode. It must provide a stable potential that is relatively invariable, despite changes in either the chemical composition or physical properties of the process stream. (3)

4.3 Hydrogen ions. (1)

4.4 0-7pH. (1)

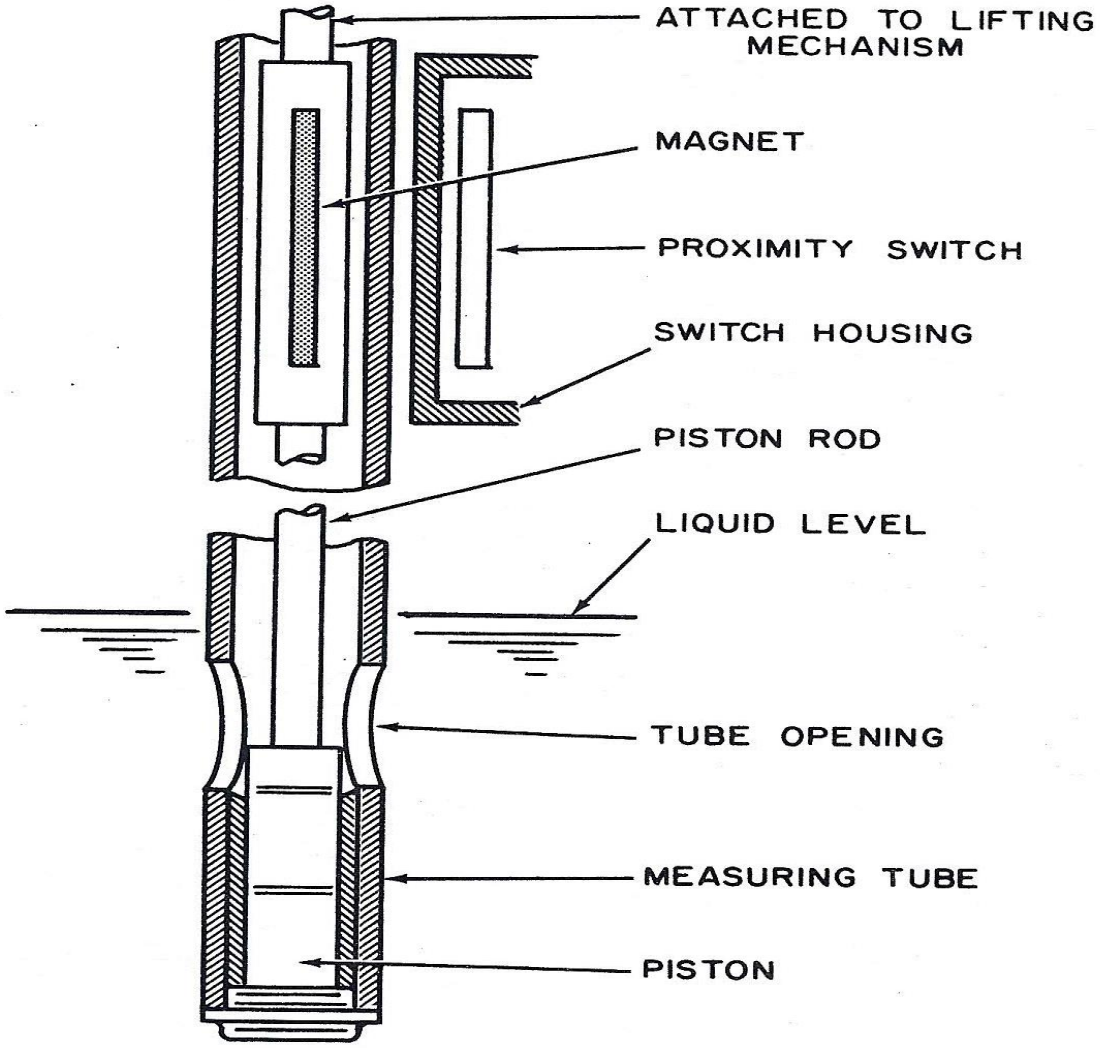
4.5 A screened glass electrode can be used and the electrode assembly must be located at a point where the residual field is smallest. (3)

[10]

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QUESTION 5

5.1



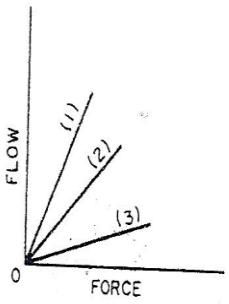
The working principle of the falling-piston viscometer is quite similar to the falling-ball viscometer. Its excellent reproducibility makes it possible to utilise it for the measurement of both Newtonian and non-Newtonian fluid viscosities as an in-process instrument.

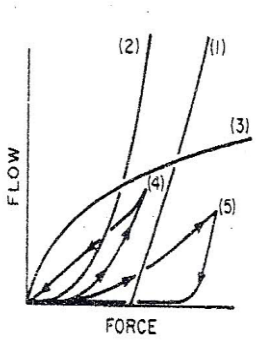
The measuring element can be installed in a tank, open or closed, or it may be in a liquid-filled line as long as the measuring tube is completely immersed in the fluid. During the filling phase, the piston at the bottom of the tube is automatically raised by an air-lifting mechanism or by a motor-cam mechanism. As the piston is raised, a sample of the liquid to be measured is drawn in through openings in the sides of the tube, and fills the measuring tube as the piston is withdrawn.

For some in-line units, the piston is raised by the process fluid. The measuring tube is filled through an opening in the bottom of the tube. The filling and measuring cycle is controlled by the periodical opening and closing of a valve. The pressure drop across the measuring tube is about 5 PSI. During the measuring phase, the piston assembly is allowed to fall by gravity, expelling the sample out of the tube through the same route that it entered. (5)

5.2 Newtonian fluids: When fluids are deformed by strain the ratio between shear rate-shear stress will be a constant value.

Non-Newtonian fluids: When fluids are deformed by strain the ratio between shear rate and shear stress will not be a constant value.



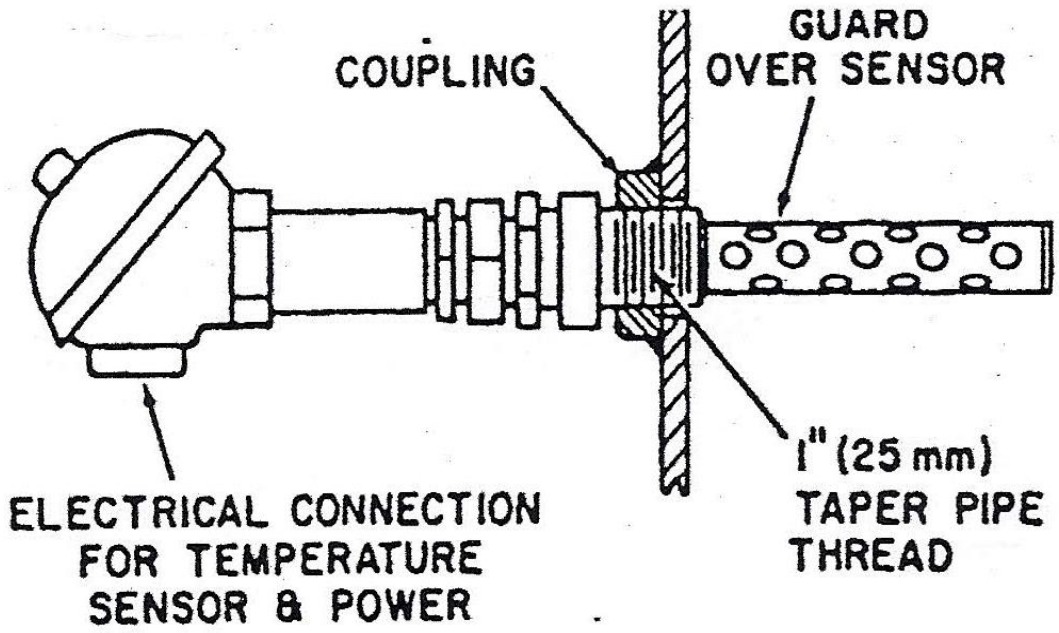


(3)
[8]

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QUESTION 6

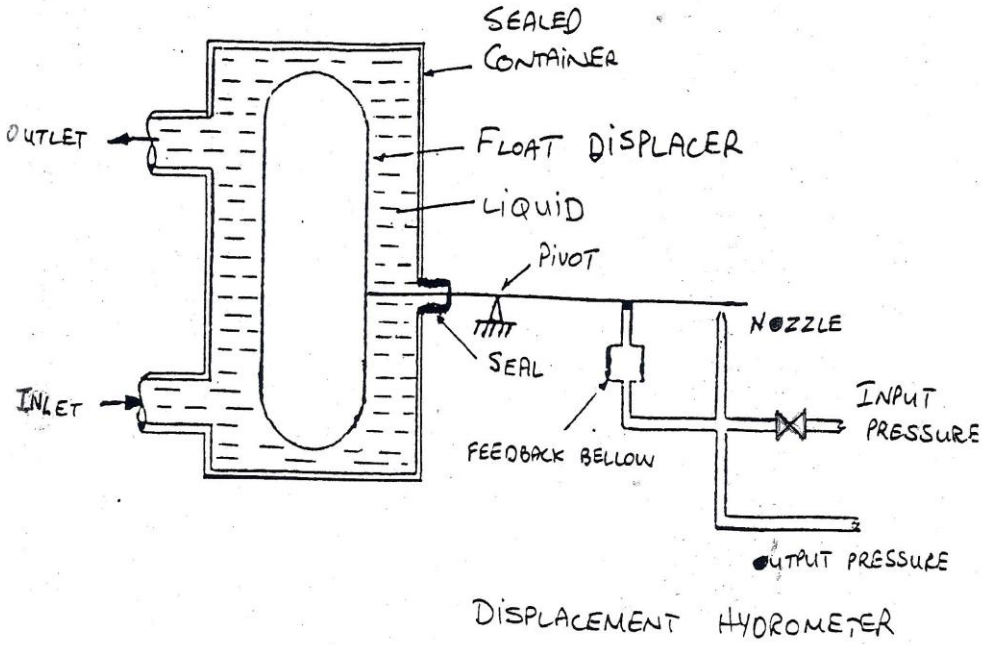
6.1



All solids condense and absorb small quantities of water on their surfaces. If the solid is soluble, a layer of saturated solution is formed of which the vapour pressure is lower than that of pure water at the same temperature. Some solids form saturated solutions in water of which the vapour pressure is less than the partial pressure of water vapour in the atmosphere contacting the solution. When that is the case, water vapour will continue to condense until the vapour pressure of the solution equals the partial pressure of the water vapour. On the other hand, water vapour will evaporate from the solution should the water-vapour partial pressure decrease below the vapour pressure of the solution. Under steady state conditions, solution vapour pressure equals water vapour partial pressure, and knowledge of the former can be used to determine the moisture content of the atmosphere.
The sensing element consists of a thin-walled, hollow-metal socket wrapped with tape filled with salt crystals (Nthium chloride). Two wires are wrapped over the tape and connected to a regulated, alternating current voltage source; the electric circuit between the wires is completed by the salt crystals. When the sensing element is exposed to the sample atmosphere, water condensing on the crystals forms an ionic solution which allows an electric current to flow between the wires. This current heats the solution and raises its vapour pressure. As more water condenses, more current flows, resulting in a further increase of the solution vapour pressure until equilibrium is reached. On decreasing moisture content of the measured gas, water evaporates from the element, decreasing the current flow and resulting in a new equilibrium at a lower solution vapour pressure. A temperature sensor inside the hollow socket is used to detect sensor temperature and to provide a signal for readout. The output calibration can be in terms of dew-point temperature, water vapour pressure or specific humidity. (8)

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6.2



DISPLACEMENT HYDROMETER

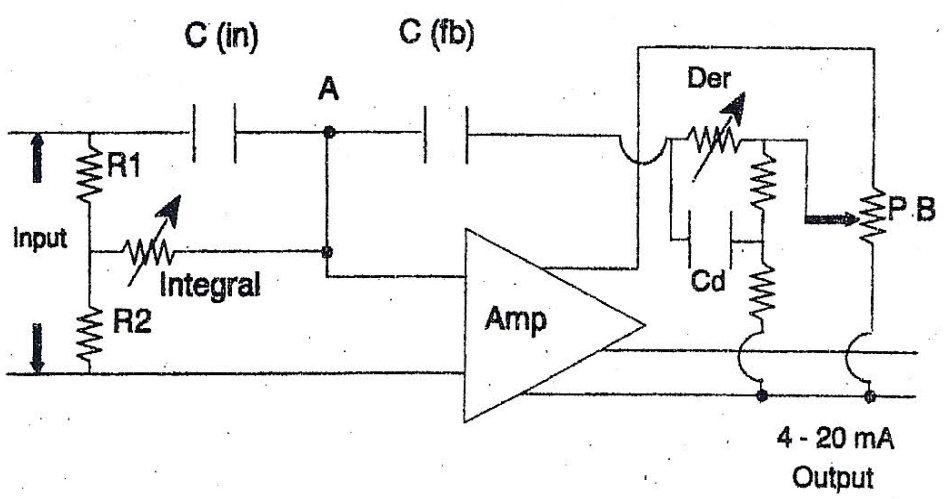
(5)
[13]

TOTAL SECTION B: 31

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SECTION C

QUESTION 7

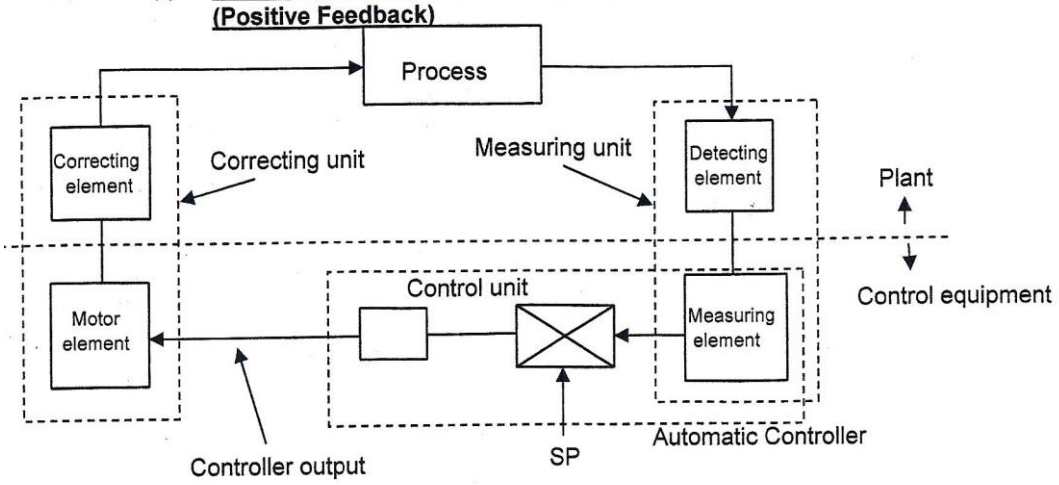


- There are two outputs from the amplifier. One comprises a current signal 4-20 mA for control and the other is a voltage signal of ±4 v for feedback purposes. These two signals are proportional to one another but are electrically insolated.
- If an error signal is impressed on the input capacitor C(in) it is charged and a potential appears at point A forming an input signal to the amplifier.
- A feedback voltage is generated to C(fb) to counteract this potential and the potential at point A returns to zero. Thus we have established a form of proportional control.
- It is necessary to have some means of adjusting the proportional band. This is shown as potentiometer Pb. For integral control the input capacitor is effectively shunted by the integral potentiometer and other resistors in the input circuit.
- It can be considered that the immediate proportional action occurs as before but the resistors provide a path for the discharge of C(in) which would mean a potential appearing at A. This would provide an input signal at the amplifier which would immediately supply a feed-back signal to counteract it and in effect recharge the capacitor.
- If the input signal is sustained the action is continued and will continue until the input signal is zero.
- Derivative control is included in the feedback path to capacitor C(fb).
- When an input signal appears there is little variation to the normal proportional action. The derivative capacitor Cd at the first instant of change transfers only a part of the voltage existing across the proportional band potentiometer Pb because of the voltage divider arrangement R1 + R2.
- As a result of this to obtain the necessary feedback to C(pb) the amplifier output is increased but as the derivative capacitor becomes charged the influence of the voltage divider is lessened and the entire voltage across Pb is supplied to the feedback capacitor C(fb).
- The voltage divider determines the amplitude of the derivative feedback. **[16]**

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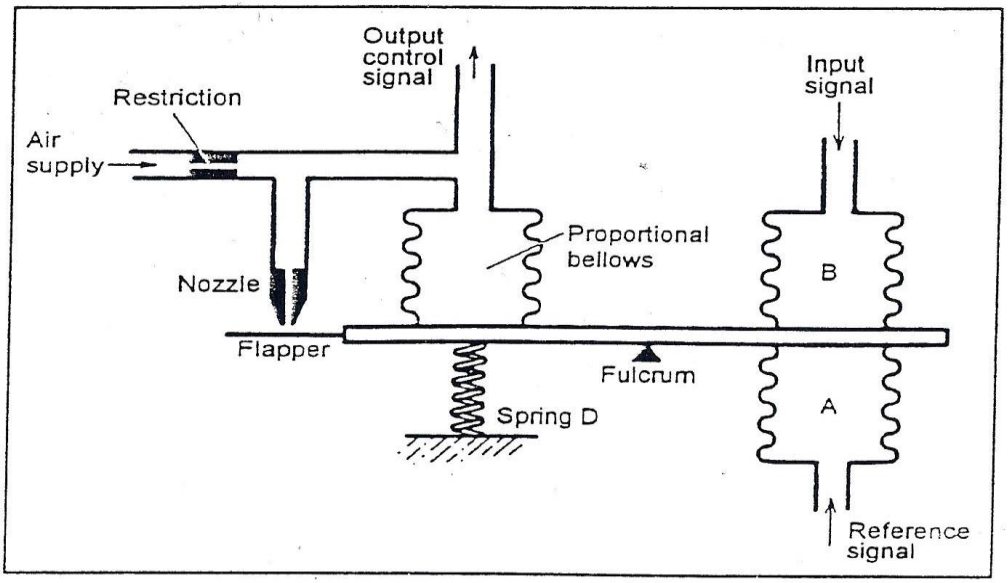
QUESTION 8

8.1 (ii) **Closed loop system** (Block diagram)
(Positive Feedback)



(10)

8.2



(5)
[15]

TOTAL SECTION C: 31
GRAND TOTAL: 100

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